

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Canon
Firmware Version: N/A
Model(s): LV-7290, LV-7295, LV-7296, LV-7390, LV-7391, LV-8225

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.00.0000-b022	2.0.0

Version History

Module Version	Date	Notes
1_0_0_0	9/13/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: InterfaceName.connectionCounter = 5

Supported Classes and Examples

SerialClass
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LV-7290')
SerialOverEthernetClass
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LV-7290')

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Normal' 'Zoom'	Value 'Full' 'True'	Value 'Wide' 'Custom'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Normal')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command ClosedCaptionDisplay	Value 'CC1' 'CC4'	Value 'CC2' 'Off'	Value 'CC3'
# ClosedCaptionDisplay example InterfaceName.Set('ClosedCaptionDisplay', 'CC1')			
Command ExecutiveMode	Value 'Off'	Value 'Remote Lock'	Value 'Key Lock'
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'Off')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'RGB' 'Scart'	Value 'Component' 'HDMI'	Value 'S-Video' 'Video'
# Input example InterfaceName.Set('Input', 'RGB')			
Command LampMode	Value 'High'	Value 'Normal'	Value 'Eco'
# LampMode example			

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InterfaceName.Set('LampMode', 'High')		
Command Power	Value 'On'	Value 'Off'
# Power example InterfaceName.Set('Power', 'On')		
Command VideoMute	Value 'On'	Value 'Off'
# VideoMute example InterfaceName.Set('VideoMute', 'On')		
Command Volume	Value 0 to 63 in steps of 1	
# Volume example InterfaceName.Set('Volume', 63)		

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus, and DeviceStatus do not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})  
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})  
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)  
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)  
Value = InterfaceName.ReadStatus(Command)  
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)  
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AspectRatio	'Normal'	'Full'	'Wide'
	'Zoom'	'True'	'Custom'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	Value
ClosedCaptionDisplay	'CC1'	'CC2'	'CC3'
	'CC4'	'Off'	
# ClosedCaptionDisplay examples InterfaceName.Update('ClosedCaptionDisplay') Value = InterfaceName.ReadStatus('ClosedCaptionDisplay') InterfaceName.SubscribeStatus('ClosedCaptionDisplay', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
DeviceStatus	'Power Failure'		

		'Cooling down due to abnormal temperature'	'Standby after Cooling Down'
	'Power saving Cooling Down'	'Power saving'	'Cooling down due to Lamp Failure'
	'Standby after Cooling Down due to Lamp Failure'	'Normal'	
# DeviceStatus examples Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command ExecutiveMode	Value 'Off'	Value 'Remote Lock'	Value 'Key Lock'
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command FilterUsage	Value 0 –		
# FilterUsage examples InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'RGB' 'Scart'	Value 'Component' 'HDMI'	Value 'S-Video' 'Video'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'High'	Value 'Normal'	Value 'Eco'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage examples InterfaceName.Update('LampUsage')			

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Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command OperationHours	Value Hours		
# OperationHours examples InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)			
Command Power	Value 'On' 'Cooling Down'	Value 'Off'	Value 'Warming Up'
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			
Command Volume	Value 0 to 63 in steps of 1		
# Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

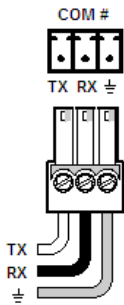
Data Bits: 8

Parity: None

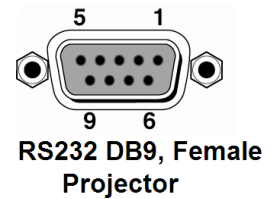
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Appendix A. Set Commands

Aspect Ratio Custom	CF SCREEN CUSTOM\x0D
Aspect Ratio Full	CF SCREEN FULL\x0D
Aspect Ratio Normal	CF SCREEN NORMAL\x0D
Aspect Ratio True	CF SCREEN TRUE\x0D
Aspect Ratio Wide	CF SCREEN WIDE\x0D
Aspect Ratio Zoom	CF SCREEN ZOOM\x0D
Audio Mute Off	CF MUTE OFF\x0D
Audio Mute On	CF MUTE ON\x0D
Auto Image None	CF AUTOSETUP START\x0D
Closed Caption Display CC1	CF CCAPTIONDISP CC1\x0D
Closed Caption Display CC2	CF CCAPTIONDISP CC2\x0D
Closed Caption Display CC3	CF CCAPTIONDISP CC3\x0D
Closed Caption Display CC4	CF CCAPTIONDISP CC4\x0D
Closed Caption Display Off	CF CCAPTIONDISP OFF\x0D
Executive Mode Key Lock	CF KEYDIS KEY\x0D
Executive Mode Off	CF KEYDIS NONE\x0D
Executive Mode Remote Lock	CF KEYDIS RC\x0D
Freeze Off	CF FREEZE OFF\x0D
Freeze On	CF FREEZE ON\x0D
Input Component	CF INPUT1 YBPR\x0D
Input HDMI	CF INPUT2 HDMI\x0D
Input RGB	CF INPUT1 ANALOG\x0D
Input Scart	CF INPUT1 SCART\x0D
Input S-Video	CF INPUT1 S-VIDEO\x0D
Input Video	CF INPUT3 VIDEO\x0D
Lamp Mode Eco	CF LAMPMODE ECO\x0D
Lamp Mode High	CF LAMPMODE HIGH\x0D
Lamp Mode Normal	CF LAMPMODE NORMAL\x0D
Power Off	CF POWER OFF\x0D
Power On	CF POWER ON\x0D
Video Mute Off	CF VMUTE OFF\x0D
Video Mute On	CF VMUTE ON\x0D
Volume 0	CF VOLUME 000\x0D
Volume 63	CF VOLUME 063\x0D

Appendix B. Update Commands

Aspect Ratio	CR SCREEN\x0D
Audio Mute	CR MUTE\x0D
Closed Caption Display	CR CCAPTIONDISP\x0D
Executive Mode	CR KEYDIS\x0D
Filter Usage	CR FILH\x0D
Freeze	CR FREEZE\x0D
Input	CR INPUT\x0D
Lamp Mode	CR LAMPMODE\x0D
Lamp Usage	CR LAMPH\x0D
Operation Hours	CR PROJH\x0D
Power	CR STATUS\x0D
Video Mute	CR VMUTE\x0D
Volume	CR VOLUME\x0D